

Module 8 — Azure Integration (Minimal & Justified)

Customer Engagement & Service Analytics Platform

Objective

This document presents a conceptual integration of Microsoft Azure services into the Customer Engagement platform. The goal is to demonstrate secure identity management using Azure Active Directory and automated serverless logic using Azure Functions, without requiring actual deployment. The design supports role-based access control, secure authentication, and proactive handling of SLA breaches.

User Story 10 — Secure Identity using Azure Active Directory (Conceptual)

Azure Active Directory (Azure AD) is conceptually integrated to provide secure authentication and role-based authorization for customers, support agents, supervisors, and administrators.

Conceptual Authentication & Authorization Flow

- User accesses the application
- Application redirects to Azure AD login
- User authenticates with credentials
- Azure AD issues authentication token
- Application validates token
- Role-based access is granted

Role-Based Access Mapping

- Customer — Submit and track tickets
- Support Agent — Manage assigned tickets
- Supervisor — Monitor operations
- Administrator — Full system access

Justification for Azure AD Usage

- Enterprise-grade security
- Single Sign-On support
- Centralized identity management

- Reduced custom authentication complexity
- Highly scalable

User Story 11 — Serverless Logic using Azure Functions (Conceptual)

Azure Functions are conceptually used to automate ticket escalation when SLA thresholds are exceeded, enabling proactive service management.

Escalation Scenario

- Monitor open tickets
- Detect SLA violations
- Trigger escalation
- Notify supervisors
- Increase ticket priority

Conceptual Pseudo-Code

For each open ticket:

If $\text{current_time} - \text{created_time} > \text{SLA_threshold}$:

 Mark as SLA breach

 Notify supervisor

 Increase priority

Cost-Aware Justification

- Pay-per-execution pricing
- No server maintenance
- Automatic scaling
- Ideal for event-driven tasks

Conclusion

The conceptual Azure integration strengthens the platform by providing secure identity management and automated SLA monitoring. Azure AD ensures controlled access through role-based authorization, while Azure Functions enable cost-effective automation without infrastructure overhead.